Matthew Badger

Curriculum Vitae

https://tangentmeasure.com/

matthew.badger@uconn.edu

Employment and Visiting Positions

University of Connecticut

- Professor, Department of Mathematics, Fall 2025 – *present*.
- Associate Professor, Department of Mathematics, Fall 2019 – Summer 2025.
- Assistant Professor, Department of Mathematics, Fall 2014 – Summer 2019.

Stony Brook University

- James H. Simons Instructor, Department of Mathematics, Fall 2011 – Spring 2014.

Visiting Positions

- PCMI, Harmonic Analysis, July 2018.
- MSRI, Harmonic Analysis, January – March 2017.
- IPAM, Interactions Between Analysis and Geometry, March – June 2013.

Education

University of Washington

- Ph.D. in Mathematics, August 2011. *Thesis advisor*: Tatiana Toro.
- M.S. in Mathematics, August 2008.

University of Pittsburgh

- B.S. in Mathematics, April 2006.
- B.A. in Political Science, April 2006.

Research Interests

My main research focus is on the **Geometry of Sets and Measures** that appear in various types of classical analysis. I am especially interested in problems from geometric measure theory and in higher-dimensional (3D+) analogues of geometric function theory in the complex plane (2D). In lay terms, two of my current threads of investigation involve (i) understanding the geometric properties of the **creases in crumpled pieces of paper**, and (ii) understanding the dimension of the active interface where random **particles of air reach the alveoli in human lungs**. In addition to analysis, my work involves elements of discrete mathematics and numerical simulation.

Buzzwords

Geometric Measure Theory, Rectifiability, Harmonic Measure, Elliptic and Parabolic PDE in Rough Domains, *Lipschitz and Hölder Parameterizations*, *Modulus of Measures*, *Quasiconformal Maps*

Research Grants, Fellowships, Awards

- NSF 2154047, "Rectifiability and Fine Geometry of Sets, Radon Measures, Harmonic Functions, and Temperatures", 2022–2025, $278,485
- **NSF 1650546, "CAREER: Analysis and Geometry of Measures", 2017–2022, $410,000**

- NSF 1500382, “Geometry of Sets and Measures”, 2015–2018, $120,000
- **NSF Mathematical Sciences Postdoctoral Research Fellowship, 2012–2015, $150,000**
- RTG Graduate Fellowship, Inverse Problems & PDE, University of Washington, 2009–2010 and 2010–2011.
- VIGRE Graduate Fellowship, University of Washington, 2006–2007.
- M.M. Culver Award, University of Pittsburgh, 2004, 2005 and 2006: prize awarded annually to outstanding undergraduates majoring in mathematics.

Conference Grants

- NSF 2403968, “Conference: Geometry of Measures and Free Boundaries, 2024–2025, $50,000 (Organization: University of Washington, PI: Bobby Wilson, co-PIs: Matthew Badger, Mariana Smit Vega Garcia, Stefan Steinerberger)
  International research conference and mini-courses held at University of Washington with approximately 80 participants from 23 U.S. states and 12 countries.
- NSF 1901256, “Collaborative Research: The Northeast Analysis Network”, 2019–2023, $10,305
  Regional conference series with rotating locations at SUNY Albany, Syracuse University, University of Connecticut, University of Rochester. This award supported the edition of the conference held at UConn in 2019.

Thesis Students

Thesis Advisor for:

- **Lisa Naples** (supervised 2016–2020), UConn, Ph.D. Summer 2020.
  Fall 2020–Spring 2023: Visiting Assistant Professor, Macalester College, St. Paul, MN.
  Fall 2023–present: Assistant Professor, Fairfield University, Connecticut.

Committee Member / Examiner for:

- **Michael Albert**, UConn, current PhD Student, expected to graduate in Spring 2026
  During academic year Fall 2023–Spring 2024, I met with Michael weekly while his advisor was on sabbatical. Michael has been my collaborator since Summer 2024.
- **Paul Tee**, UConn, current PhD student, expected to graduate in Spring 2026
- **E Koenig**, University of Minnesota, current PhD Student, passed oral exam in Spring 2025
- **Timo Takala**, Aalto University, Ph.D. Summer 2025
  I served as the opponent on Timo’s PhD defense.
- **Pablo Hidalgo Palencia**, Universidad Complutense de Madrid, Spring 2025
  I was a pre-defense evaluator for Pablo’s thesis.
- **Eve Shaw**, The University of Tennessee, Ph.D. Spring 2025
  Fall 2025–present: J.L. Doob Research Assistant Professor, UIUC
- **Polina Perstneva**, University of Paris-Saclay, Ph.D. December 2023
- **Marco Carfagnini**, UConn, Ph.D. Summer 2022.
  Fall 2022–Spring 2024: Stefan E. Warschawski Visiting Assistant Professor, UCSD Fall 2024–present: Assistant Professor, University of Melbourne, Australia

Graduate Research Assistants Funded

- **Corrie Ingall**, UConn, Spring 2025

- **Alec Wendland**, UConn, Summer 2024
- **Michael Albert**, UConn, Spring 2024 and Summer 2025
- **Zackary Boone**, UConn, Summer 2023
- **Christopher Hayes**, UConn, Fall 2021
- **Erik Wendt**, UConn, Fall 2020
- **Lisa Naples**, UConn, Fall 2017, Fall 2018, Fall 2019

Visiting Ph.D. Students Hosted for Extended Visits

- **Jared Krandel**, Stony Brook University, Spring 2023 (2 weeks)
- **Cole Jeznach**, University of Minnesota, Fall 2022 (2 weeks)
- **Sean McCurdy**, University of Washington, Fall 2018 (3 months)

Postdoctoral Fellows

- **Alyssa Genschaw**, Visiting Assistant Professor at UConn, 2019–2021. Since Fall 2021, appointed Assistant Professor at Milwaukee School of Engineering. Currently under consideration for promotion to Associate Professor.
- **Murat Akman**, Evarist Gine Postdoctoral Fellow at UConn, 2016–2019. Awarded AMS-Simons travel grant in 2018. In Fall 2019, appointed Lecturer at University of Essex, UK. Promoted to Senior Lecturer.
- **Vyron Vellis**, Postdoctoral Fellow at UConn, 2016–2019. Awarded NSF grant as a postdoc at UConn in 2018. In Fall 2019, appointed to Assistant Professor at University of Tennessee. Promoted and tenured as Associate Professor in Fall 2023.

Additional Current Mentoring

- In Spring 2026, I am directing a reading course on local set approximation with two first year math Ph.D. students at UConn. After the students learn sufficient background, I intend for us to work on a joint research project.
- CAPS research mentor for **Kevin Palmer** a junior math major at UConn. We meet weekly and I am helping Kevin to prepare to submit a proposal for the NSF GRF in Fall 2026.

Publications *(ordered by arXiv number / date of first availability)*

1. Harmonic polynomials and tangent measures of harmonic measure, *Rev. Mat. Iberoam.* **27** (2011), no. 3, 841–870. arXiv:0910.2591
2. Null sets of harmonic measure on NTA domains: Lipschitz approximation revisited, *Math. Z.* **270** (2012), no. 1–2, 241–262. arXiv:1003.4547
3. Flat points in zero sets of harmonic polynomials and harmonic measure from two sides, *J. London Math. Soc.* **87** (2013), no. 1, 111–137. arXiv:1109.1427
4. (with James T. Gill, Steffen Rohde and Tatiana Toro) Quasisymmetry and rectifiability of quasispheres, *Trans. Amer. Math. Soc.* **366** (2014), no. 3, 1413–1431. arXiv:1201.1581
5. Beurling's criterion and extremal metrics for Fuglede modulus, *Ann. Acad. Sci. Fenn. Math.* **38** (2013), 677–689. arXiv:1207.5277
6. (with Raanan Schul) Multiscale analysis of 1-rectifiable measures: necessary conditions, *Math. Ann.* **361** (2015), no. 3–4, 1155–1172. arXiv:1307.0804
7. (with Jonas Azzam and Tatiana Toro) Quasiconformal planes with bi-Lipschitz pieces and extensions of almost affine maps, *Adv. Math.* **275** (2015), 195–259. arXiv:1403.2991

8. (with Stephen Lewis) Local set approximation: Mattila-Vuorinen type sets, Reifenberg type sets, and tangent sets, *Forum Math. Sigma* **3** (2015), e24, 63 pp. arXiv:1409.7851
9. (with Raanan Schul) Two sufficient conditions for rectifiable measures, *Proc. Amer. Math. Soc.* **144** (2016), no. 6, 2445-2454. arXiv:1412.8357
10. (with Murat Akman, Steve Hofmann, and José María Martell) Rectifiability and elliptic measures on 1-sided NTA domains with Ahlfors-David regular boundaries. *Trans. Amer. Math. Soc.* **369** (2017), no. 8, 5711–5745. arXiv:1507.02039
11. (with Max Engelstein and Tatiana Toro) Structure of sets which are well approximated by zero sets of harmonic polynomials, *Anal. PDE.* **10** (2017), no. 6, 1455–1495. arXiv:1509.03211
12. (with Raanan Schul) Multiscale analysis of 1-rectifiable measures II: characterizations. *Anal. Geom. Metr. Spaces* **5** (2017), no. 1, 1–39. arXiv:1602.03823
13. (with Vyron Vellis) Geometry of measures in real dimensions via Hölder parameterizations. *J. Geom. Anal.* **29** (2019), no. 2, 1153–1192. arXiv:1706.07846
14. Generalized rectifiability of measures and the identification problem. *Complex Analysis and its Synergies* **5** (2019), 2. doi:10.1007/s40627-019-0027-3 arXiv:1803.10022
15. (with Lisa Naples and Vyron Vellis) Hölder curves and parameterizations in the Analyst's Traveling Salesman theorem. *Adv. Math.* **349** (2019), 564–647. doi:10.1016/j.aim.2019.04.011 arXiv:1806.01197
16. (with Max Engelstein, Tatiana Toro) Regularity of the singular set in a two-phase problem for harmonic measure with Hölder data. *Rev. Mat. Iberoam.* **36** (2020), no. 5, 1375–1408. doi:10.4171/rmi/1170 arXiv:1807.08002
17. (with Vyron Vellis) Hölder parameterization of iterated function systems and a self-affine phenomenon. *Anal. Geom. Metr. Spaces* **9** (2021), 90–119. doi:10.1515/agms-2020-0125 arXiv:1910.08850
18. (with Sean McCurdy) Subsets of rectifiable curves in Banach spaces I: sharp exponents in traveling salesman theorems. *Illinois J. Math.* **67** (2023), no. 2, 203–274. doi:10.1215/00192082-10592363 arXiv:2002.11878v3
19. (with Lisa Naples) Radon measures and Lipschitz graphs. *Bull. London Math. Soc.* **53** (2021), no. 3, 921–936. doi:10.1112/blms.12473 arXiv:2007.08503
20. (with Alyssa Genschaw) Hausdorff dimension of caloric measure. *Amer. J. Math.* **147** (2025), no. 2, 465–502. doi:10.1353/ajm.2025.a954648 arXiv:2108.12340
21. (with Sean Li and Scott Zimmerman) Identifying 1-rectifiable measures in Carnot groups. *Anal. Geom. Metr. Spaces*. **11** (2023), Paper no. 20230102, 40 pp. doi:10.1515/agms-2023-0102 arXiv:2109.06753v4
22. (with Sean McCurdy) Subsets of rectifiable curves in Banach spaces II: universal estimates for almost flat arcs. *Illinois J. Math.* **67** (2023), no. 2, 275–331. doi:10.1215/00192082-10592390 arXiv:2208.10288
23. (with Alyssa Genschaw) Lower bounds on Bourgain's constant for harmonic measure. *Canad. J. Math.* **76** (2024), no. 6, 1967–1986. doi:10.4153/S0008414X2300069X arXiv:2205.15101v2
24. (with Max Engelstein and Tatiana Toro) Slowly vanishing mean oscillations: non-uniqueness of blow-ups in a two-phase free boundary problem. *Vietnam J. Math.* **52** (2024), 615–625. doi:10.1007/s10013-023-00668-6 arXiv:2210.17531
25. A practical guide to writing an NSF grant proposal, *Notices Amer. Math. Soc.* **70** (2023), no. 7, 1089–1093. doi:10.1090/noti2729 Click for PDF
    Note: This article aimed at graduate students and junior faculty was written after I was contacted by the editors of Early Career Section of the AMS Notices.

26. (with Raanan Schul) Square packings and rectifiable doubling measures. *Discrete Anal.* **2025**:3, 40pp. Link to Article arXiv:2309.01283
27. (with Cole Jeznach) On the number of nodal domains of homogeneous caloric polynomials. *Annales Henri Lebesgue* **9** (2026), 261–298. doi:10.5802/ahl.266 arXiv:2401.07268v2

Preprints

28. (with Jared Krandel and Vyron Vellis) Tangents to Lipschitz and Sobolev images. Preprint. arXiv:2601.22473

In Preparation

29. (with Michael Albert and Alyssa Genschaw) One-ring Bourgain alternative and a 6-decimal bound on the dimension of harmonic measure.

Plenary Talks

- "Square Packings, Analysis, and Metric Geometry". **Spring 2025 Taft Lecture**. University of Cincinnati. Cincinnati, Ohio. April 2025.
- "Open Problems about Curves, Sets, and Measures". **8th Ohio River Analysis Meeting**. Lexington, Kentucky. March 2018.

Other Invited Research Talks *(reverse chronological order)*, 85 invitations

Upcoming Invited Talks

- IRP Workshop on Free Boundary Problems. Centre de Recerca Matemàtica (CRM). Universitat Autònoma de Barcelona. Bellaterra, **Spain**. November 2026.
- Philadelphia Harmonic Analysis and Differential Equations. **2026 ICM Official Sattelite Conference.** Philadelphia, Pennsylvania. July 2026.

Past Invited Talks

- Colloquium. Stony Brook University. April 2026.
- AMS Special Session on "Analysis, Probability, and Geometric Mapping Theory in Non-Smooth Spaces". Eastern Section. Boston, Massachusetts. March 2026.
- Workshop / Winter School on Minimal Surfaces and Rectifiability. Institute for Theoretical Sciences. Westlake University. Hangzhou, **China**. January 2026. (Declined Invitation.)
- AMS Special Session on "Harmonic Analysis and Elliptic PDE". Joint Mathematics Meetings. Washington, District of Columbia. January 2026.
- Colloquium. University of Pittsburgh. Pittsburgh, Pennsylvania. October 2025
- Workshop on Regularity Theory of PDEs and Calculus of Variations 2025. Essex, **United Kingdom**. June 2025.
- Quasiweekend III—twenty years on. Helsinki, **Finland**. June 2025. This is the third edition of a conference series in metric geometry and quasiconformal analysis, which takes place once every ten years.
- AMS Special Session on "Analysis, Geometry, and Dynamics in Metric Spaces". Eastern Section. Hartford, Connecticut. April 2025.
- AMS Special Session on "Non-smooth analysis and geometry". Joint Mathematics Meetings. Seattle, Washington. January 2025.
- International Conference on Special Functions, Analytic Functions, Metrics and Quasiconformality. Indore, **India**. December 2024. (Declined Invitation.)
- Conference. "PDE in Moab". Utah. June 2024.

- Conference. University of Arkansas Spring Lecture Series. Fayetteville, AR. May 2024.
- AMS Special Session. Interactions between Analysis, Geometric Measure Theory, and Probability in Non-smooth Spaces. Howard University. Washington, DC. April 2024.
- Colloquium. University of Tennessee. Knoxville, TN. February 2024.
- Analysis seminar. University of Tennessee. Knoxville, TN. February 2024.
- Harmonic analysis seminar. University of Paris-Saclay. Orsay, **France**. December 2023.
- PDE Seminar. University of Minnesota. Minneapolis, MN. October 2023.
- Calderón-Zygmund Seminar. University of Chicago. Chicago, Illinois. January 2023.
- Math Seminar. Montana State University. Bozeman, Montana. October 2022.
- AMS special session on Nonsmooth Analysis and Geometry. Eastern section. Amherst, Massachusetts. October 2022.
- Analysis and Probability Seminar. UConn. Storrs, Connecticut. September 2022.
- Conference. Geometry and Analysis on Nonsmooth Spaces. Texas A&M. August 2022.
- CBMS conference: Analysis, Geometry, and PDEs in a Lower-Dimensional World. FSU. Tallahassee, Florida. May 2022 (Originally August 2020, Delayed due to Covid-19)
- Harmonic analysis and PDE conference. Yonsei Unviersity. Seoul, South Korea. May 2022. (Originally scheduled May 2020. Delayed due to Covid-19. Declined invitation.)
- AMS Special Session on Geometric Measure Theory. J. Math. Meet. Online. April 2022. (Originally scheduled January 2022. Delayed due to Covid-19. Declined invitation.)
- Calderón-Zygmund Seminar. University of Chicago. Virtual. May 2021.
- AMS Special Session on Nonsmooth Analysis and Geometry. Virtual Central Sectional Meeting. April 2021.
- Elliptic PDEs and Geometric Variational Problems. 13th AIMS conference on Dyn. Sys., Diff. Eqs. and Appl. Atlanta, Georgia. June 2020. (Cancelled due to Covid-19)
- BIRS Workshop. Analysis and Geometry of Metric Spaces - a Bridge between Smooth and Fractal Views. Banff, Canada. April 2020. (Cancelled due to Covid-19)
- AMS special session on Harmonic Analysis. Southeastern Sectional Meeting. Charlottesville, Virginia. March 2020. (Cancelled due to Covid-19)
- Special Session on Interfaces between PDEs and Geometric Measure Theory. Joint Mathematics Meetings. Denver, Colorado. January 2020.
- Colloquium. Kansas State University. Manhattan, Kansas. November 2019.
- Function theory seminar. Kansas State University, Manhattan, Kansas. November 2019.
- Geometry and analysis in function and mapping theory on Euclidean and metric measure spaces. Banach Center. **Poland**. October 2019.
- AMS special session on Analysis and Probability on Metric Spaces and Fractals. Central Sectional Meeting. Madison, Wisconsin. September 2019.
- Rainwater seminar. University of Washington. Seattle, Washington. February 2019.
- AMS special session on Harmonic Analysis, Partial Differential Equations, and Applications. Joint Mathematics Meeting. Baltimore, Maryland. January 2019.
- AMS special session on Harmonic Analysis and PDE. Southeastern Sectional Meeting. Fayetteville, Arkansas. November 2018.
- Research program seminar. PCMI Harmonic Analysis. Park City, Utah. July 2018.
- Rainwater seminar. Unviersity of Washington, Seattle, Washington. May 2018.
- AMS special session on Analysis and Geometry in Non-smooth Spaces. Eastern Sectional Meeting. Boston, Massachusetts. April 2018.

- AMS special session on Regularity of PDEs in Rough Domains. Eastern Sectional Meeting. Boston, Massachusetts. April 2018.
- AMS special session on Analysis at the Intersection of Geometric Measure Theory and Partial Differential Equations. Western Sectional Meeting. Portland, Oregon. April 2018.
- Analysis and PDE seminar. University of Kentucky. Lexington, Kentucky. March 2018.
- 3rd Northeast Analysis Network Meeting. Syracuse, New York. September 2017.
- Geometric Measure Theory conference. Warwick, **United Kingdom**. July 2017.
- Harmonic analysis seminar. MSRI. Berkeley, California. March 2017.
- Conference on “Analysis on Metric Spaces.” University of Pittsburgh. March 2017.
- Analysis seminar. Brown University. Providence, RI. October 2016.
- Analysis and PDE seminar. University of Cincinnati. Cincinnati, Ohio. April 2016.
- Rainwater seminar. University of Washington. Seattle, Washington. March 2016.
- Analysis and probability seminar. University of Connecticut. Storrs. March 2016.
- AMS Special Session on Analysis and Geometry in Nonsmooth Metric Measure Spaces. Joint Mathematics Meetings. Seattle, Washington. January 2016.
- Minisymposiam, “New Trends in Elliptic and Parabolic PDEs” SIAM conference—Analysis of PDE. Scottsdale, Arizona. December 2015.
- Rainwater seminar. University of Washington. Seattle, Washington. May 2015.
- Analysis seminar. Brown University. Providence, Rhode Island. April 2015.
- Analysis seminar. University of Missouri. Columbia, Missouri. April 2015.
- Colloquium. Worcester Polytechnic Institute. Worcester, Massachusetts. March 2015.
- Geometry seminar. University of Jyväskylä. Jyväskylä, **Finland**. February 2015.
- Geometric analysis seminar. University of Helsinki. Helsinki, **Finland**. February 2015.
- Analysis seminar. Fordham University. Bronx, New York. November 2014.
- Ahlfors-Bers Colloquium VI. Yale University. New Haven, Connecticut. October 2014.
- Analysis and probability seminar. University of Connecticut. Storrs. September 2014.
- Dynamical systems seminar. Stony Brook University and the Institute for Mathematical Sciences. Stony Brook, New York. May 2014.
- Colloquium. University of Connecticut. Storrs, Connecticut. January 2014.
- Rainwater seminar. University of Washington. Seattle, Washington. January 2014.
- Analysis seminar. University of Illinois, Urbana-Champaign. November 2013.
- Analysis seminar. Syracuse University. Syracuse, New York. October 2013.
- AMS Special Session on Harmonic Analysis and Partial Differential Equations. Louisville, Kentucky. October 2013.
- AMS Special Session on Analysis, Dynamics and Geometry in and around Teichmüller spaces. Ames, Iowa. April 2013.
- Analysis seminar. Temple University. Philadelphia, Pennsylvania. February 2013.
- Complex analysis seminar. University of Washington. Seattle. January 2013.
- Perspectives in HA, GMT, and PDE, and their applications to SCV. Temple University. Philadelphia, Pennsylvania. September 2012.
- Simons Postdoctoral Fellows Meeting. Simons Center for Geometry and Physics. Stony Brook, New York. April 2012.
- Complex analysis seminar. University of Washington. Seattle. January 2012.
- Dynamical systems seminar. Stony Brook University and the Institute for Mathematical Sciences. Stony Brook, New York. September 2011.

- AMS Special Session on Geometric Aspects of Analysis and Measure Theory. Eastern Sectional Meeting. Ithaca, New York. September 2011.
- AMS Special Session on Harmonic Analysis and PDE. Joint Mathematics Meetings. New Orleans, Louisiana. January 2011.
- Analysis seminar. Brown University. Providence, RI. December 2010.
- Harmonic analysis seminar. Université Paris-Sud XI. Orsay, **France**. April 2010.
- Rainwater seminar. University of Washington. Seattle, Washington. February 2010.
- Harmonic analysis conference. Centre de Recerca Matemàtica (CRM). Bellaterra, **Spain**. June 2009.
- Rainwater seminar. University of Washington. Seattle, Washington. May 2009.

Invited Mini Courses

- *Introduction to Harmonic and Caloric Measure*, 3 lectures, 3 hours. Geometry of Measures and Free Boundaries. University of Washington. July 2024.
- *Introduction to Geometry of Sets and Measures*, 3 lectures, $4\frac{1}{2}$ hours. Progress in Harmonic Analysis and Geometric Measure Theory. Temple University. April 2014.

Professional Activities

- Co-organizer (with Ryan Alvarado and Lisa Naples): AMS Special Session in Nonsmooth Analysis and Geometry. Fall Eastern Section Meeting. SUNY Albany. October 2024.
- Co-organizer of the 2024 Northeast Analysis Network Conference at Syracuse University. September 2024. *This regional conference series is a joint venture with SUNY Albany, Syracuse University, and University of Rochester.*
- Organizer: Geometry of Measures and Free Boundaries. A conference in honor of Tatiana Toro. Seattle, Washington. July 22–26, 2024. With four Introductory Mini-Courses held before the conference July 20–21, 2024. Co-organized with Mariana Smit Vega Garcia, Stefan Steinerberger, Bobby Wilson. *This international research conference attracted about 75 mathematicians, including speakers from 18 US States and from Australia, Finland, France, Italy, Mexico, Taiwan, United Kingdom*
- Co-organizer of the 2023 Northeast Analysis Network Conference at University of Rochester. September 2023. *This regional conference series is a joint venture with SUNY Albany, Syracuse University, and University of Rochester.*
- Co-organizer of the 2022 Northeast Analysis Network Conference at SUNY Albany. September 2022. *This regional conference series is a joint venture with SUNY Albany, Syracuse University, and University of Rochester.*
- Co-organizer (with Guy C. David and Lisa Naples): AMS Special Session in Geometry of Measures and Metric Spaces. Spring Central Section Meeting. Online. March 2022. (Originally at Purdue University, Covid-19 impacted)
- Co-organizer and Local Organizer of the 2019 Northeast Analysis Network Conference at UConn Storrs Campus. September 2019. *This regional conference series is a joint venture with SUNY Albany, Syracuse University, and University of Rochester.*
- Local organizer for AMS Eastern Sectional Meeting. Hartford, Connecticut. April 2019. *As the Local Organizer, I ensured smooth logistics at the UConn Hartford conference site, securing space for 20+ parallel sessions and plenary talks, 400+ participants, catering, and facilitating communication and coordination of staff from UConn and the American Mathematical Society. Planning started over 18 months in advance of the event.*

- Organizer of “Geometric and Harmonic Analysis: a Conference for Graduate Students”. University of Connecticut. March 2019. Co-organized with Murat Akman, Vyron Vellis. Website at: `https://gaha2019.math.uconn.edu/`
- Organizer of “Nonsmooth Analysis: a Workshop for Postdocs”. University of Connecticut. November 2017. Website at: `https://nonsmooth2017.math.uconn.edu/`
- Co-organizer (with Vasileios Chousionis): AMS Special Session in Geometric Aspects of Harmonic Analysis. Fall Eastern Sectional Meeting. Brunswick, Maine. September 2016.
- Co-organizer (with Chris Bishop and Raanan Schul): AMS Special Session in Geometric Measure Theory and Its Applications. Spring Eastern Sectional Meeting. Stony Brook, New York. March 2016.
- Chair of organizing committee for the UConn Special Semester in Nonsmooth Analysis. University of Connecticut. Department of Mathematics. Fall 2015. Co-organized with Vasileios Chousionis, Masha Gordina, Luke Rogers, and Sasha Teplyaev.
- Co-organizer (with Jim Gill and Joan Lind): AMS Special Session in Complex Analysis, Probability and Metric Geometry. Spring Southeastern Sectional Meeting. Knoxville, Tennessee. March 2014.
- Co-organizer (with Theresa Anderson, Nathan Pennington, Eric Stachura): AMS Special Session in Harmonic Analysis, PDE, and Geometric Measure Theory. Joint Mathematics Meetings. San Diego. January 2013.
- TA for Summer Graduate Courses in Geometric Measure Theory. MSRI. July 2011.
- Organized a series of “Professional Development Sessions” for the mathematics graduate students at University of Washington, 2010 – 2011.
- Served on **5 NSF Review Panels** and as an ad-hoc reviewer.
- Ad-hoc reviewer for European Research Council.
- Reviewed book manuscripts for Springer.
- Reviewer for *Mathematical Reviews*, 27 reviews, 2012–2017.
- Refereed papers for multiple research journals, including
  ▷ Adv. Math. ▷ Amer. J. Math. ▷ Amer. Math. Monthly ▷ Ann. Acad. Sci. Fenn. Math.
  ▷ Ann. Func.Anal. ▷ **Annals of Mathematics** ▷ Anal. PDE
  ▷ Ann. Sci. Éc. Norm. Supér. ▷ Arch. Math. (Basel) ▷ Chaos Solitons & Fractals
  ▷ **Duke Math. J.** ▷ Geom. Func. Anal. ▷ Forum Math. Sigma
  ▷ Int. Math. Res. Not. IMRN ▷ **Invent. Math.** ▷ Israel J. Math. ▷ **J. Amer. Math. Soc.**
  ▷ J. Anal. Math. ▷ **J. Eur. Math. Soc. (JEMS)** ▷ J. Fourier Anal. Appl. ▷ J. Func. Anal.
  ▷ J. Geom. Anal. ▷ J. London Math. Soc. ▷ Mathematika ▷ Math. Ann.
  ▷ Michigan Math. J. ▷ Memoirs Amer. Math. Soc. ▷ Nonlinear Anal.
  ▷ Proc. Amer. Math. Soc. ▷ Proc. London Math. Soc. ▷ Rev. Mat. Iberoam.
  ▷ Trans. Amer. Math. Soc. ▷ Vietnam J. Math.
- 80 Letters of recommendation or evaluation for: undergraduate REUs, graduate programs, postdoctoral positions, teaching-line positions, tenure-line positions, promotions, awards.

Selected Service

University of Connecticut

- CCC+ / General Education Oversight Committee, Subcommittee on Q (Quantitative) Courses, Fall 2021 – *present*
- Working Group on Quantitative Competency Standards, Fall 2023 – Spring 2024
  We revised the standards for Q Courses and added learning objectives. Now implemented.

- Reviewer for OVPR Research Excellence Program, Spring 2024

UConn College of Liberal Arts and Sciences

- Participated in CLAS Undergraduate Commencement, Spring 2016, Spring 2019, Spring 2025

UConn Department of Mathematics

- Analysis Learning Seminar, Founder and Organizer, multiple semesters
- Analysis & Probability Seminar, Organizer, multiple semesters
- Initial advisor for incoming Ph.D. students, multiple years
- Wrote and/or graded preliminary exams, multiple years
- Math department representative at college and university recruitment events for perspective and incoming undergraduate students, Fall 2023, Spring 2024
- Graduate Admissions Committee, Spring 2016, Spring 2017, Spring 2020, Spring 2023
- Graduate Admissions Co-Chair, Spring 2018 (Co-chaired with Liang Xiao)
- Graduate Program Committee, Fall 2015 – Spring 2018
- Picnic Committee, Fall 2015 – Spring 2016
- VAP Search Committee, Spring 2015

Teaching Experience

University of Connecticut

Graduate Courses

- Math 5110 Introduction to Modern Analysis **Fall 2025**
- Math 5111 Measure Theory & Integration **Spring 2025**
- Math 5440 Partial Differential Equations **Fall 2023**
- Math 5140 Fourier Analysis **Fall 2022**
- Math 5120 Complex Analysis **Spring 2020**
- Math 5010 Topics in Analysis: Geometric Measure Theory **Fall 2018**
- Math 5111 Measure Theory & Integration **Spring 2018**
- Math 5120 Complex Analysis **Spring 2016**
- Math 5110 Introduction to Modern Analysis **Fall 2015**
- Math 5110 Introduction to Modern Analysis **Fall 2014**

Undergraduate Courses – Math Majors

- Math 3151 Analysis II **Spring 2026**
- Math 3151 Analysis II **Spring 2024**
- Math 3151 Analysis II **Spring 2022**
- Math 3370 Differential Geometry **Fall 2021**
- Math 3150 Analysis I **Fall 2021**
- Math 3210 Abstract Linear Algebra **Spring 2021**
- Math 2710W Transitions to Advanced Mathematics **Spring 2019**
- Math 3151 Analysis II **Spring 2019**
- Math 3150 Analysis I **Fall 2017**
- Math 3150 Analysis I **Fall 2016**
- Math 4110 Introduction to Modern Analysis **Fall 2015**
- Math 4110 Introduction to Modern Analysis **Fall 2014**

Undergraduate Courses – Service Courses

- Math 1131Q Calculus I (2 sections) **Fall 2026**
- Math 2210Q Applied Linear Algebra (Honors) **Fall 2025**
- Math 2210Q Applied Linear Algebra (2 sections) **Fall 2024**
- Math 2210Q Applied Linear Algebra **Fall 2023**
- Math 2210Q Applied Linear Algebra (2 sections) **Spring 2023**
- Math 2210Q Applied Linear Algebra (Honors) **Spring 2021**
- Math 2210Q Applied Linear Algebra (2 sections) **Fall 2019**
- Math 2410Q Elementary Differential Equations **Fall 2016**
- Math 2410Q Elementary Differential Equations **Spring 2016**
- Math 1131Q Calculus I **Summer 2015**

Stony Brook University

Graduate Courses

- Math 550 Real Analysis II **Spring 2014**
- Math 638 Topics in Real Analysis: Brownian Motion and Harmonic Measure **Fall 2013**

Undergraduate Courses – Math Majors

- Math 322 Analysis in Several Dimensions **Spring 2012**

Undergraduate Courses — Service Courses

- Math 211 Introduction to Linear Algebra **Fall 2012**
- Math 125 Calculus A (Differential Calculus) **Fall 2011**
- Math 126 Calculus B (Integral Calculus) **Fall 2011**

In Fall 2012, I also directed two independent studies by undergraduate math majors on
▷ Harmonic Function Theory ▷ Analysis on Manifolds.

University of Washington

Undergraduate Courses — Service Courses

- Math 309 Linear Analysis **Fall 2010**
- Math 308 Matrix Algebra with Applications **Winter 2010**
- Math 308 Matrix Algebra with Applications **Summer 2008**

Grader

Graded coursework for graduate and senior undergraduate courses, held weekly office hours.

- Math 524-525-526 Real Analysis **Fall 2008 - Spring 2009**
- Math 424-425-426 Fundamental Concepts of Analysis **Fall 2007 - Spring 2008**

Teaching Assistant

Led small group sections twice a week, held weekly office hours, graded exams.

- Math 124 Calculus I **Fall 2006**

University of Pittsburgh

Teaching Assistant

Designed, presented and graded five MATLAB "exploration assignments" for sophomore level course in linear algebra required for engineering students.

- Math 0280 Introduction to Matrices and Linear Algebra **Spring 2006**

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